

brachytherapy was excluded. Three of the cohorts were from prospective clinical trials (MARS, HDR monotherapy and SPARE). For the MR-HDR boost cohort, pertinent demographic, clinical and dosimetric information were extracted from patients' medical records. Descriptive statistics were used to summarize baseline data. Toxicities experienced were coded using the Common Terminology Criteria for Adverse Events (CTCAE) version 3.0. The Nelson-Aalen estimator method using 95% confidence intervals was used to assess the cumulative incidence of biochemical failure (BCF; using the Phoenix Failure definition) and the cumulative incidences of Grade 2 or higher toxicity.

Results: One hundred forty-four patients with a mean age (65.6±6.7 years), follow-up (55.7±15.0 months) and baseline PSA (9.7±8.3 ng/mL), with predominantly cT2a (32.6%) and Gleason 7 (3+4; 62.0%) disease, were included in the final analysis. Fifteen patients treated with MR-HDR boost were followed elsewhere so no toxicity data were available at time of analysis. All risk groups were represented: low (10.4%), favourable (27.8%) and unfavourable intermediate (36.8%), and high (25.0%). Twenty-five of 137 (18%) experienced BCF with cumulative incidence rates of 20.6% at five years. A total of 41/129 (32%) patients experienced any Grade ≥2 GU (one G3) and seven of 129 patients (5.4%) experienced any Grade ≥2 GI (0 G3) toxicity. The cumulative incidence rates at five years was 33.2% for GU and 5.7% for GI, respectively. Neither BCF or toxicity (GU or GI) was significantly different between treatment groups.

Conclusions: Dose escalation to the DIL using mp-MRI dose painted HDR brachytherapy appears to be feasible, safe and well-tolerated. Further follow-up and investigation is warranted to determine whether MR-HDR improves outcomes compared to standard non MR-HDR brachytherapy.

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ULTRA-HYPOFRACTIONATED (UHF) COMPARED TO MODERATE-HYPOFRACTIONATED (MHF) PROSTATE IGRT WITH HDR BRACHYTHERAPY BOOST(BB): FOUR-YEAR TOXICITIES AND LOCAL CONTROL

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Purpose: To analyze the long-term efficacy and safety of an UHF IGRT prostate treatment regimen with HDR BB.

Materials and Methods: In this single arm, prospective monocentric study, 28 patients with median age 69 years and D'Amico's intermediate risk prostate cancer (T1c, T2a-b), were recruited between the years 2015-2016 to an experimental treatment arm of 25Gy IGRT in 5 fractions with a 15Gy HDR BB. They were then compared to two control groups, treated with either 36Gy in 12 fractions or 37.5Gy in 15 fractions with a similar HDR BB. The control groups included 151 and 311 patients respectively.

Results: Patients' characteristics were similar between the three groups for age, Gleason score, stage and initial PSA. Patients outcomes were reported using the IPSS questionnaire at baseline and at each follow-up visit until 48 months. The three groups showed a significant decrease in average IPSS over time as compared to one-month post treatment. At 12 months, the average IPSS scores were 7, 9 and 9 and drop to 5, 7 and 7 at 48 months for the 25/5, 36/12 and 37, 5/5 regimens respectively. A greater reduction was seen in the experimental arm while being non-significant. The median treatment time to deliver IGRT was six days in the UHF group compared to 16 and 21 days in the control groups. Local control rates for each group were 100%, 95% and 91% respectively. No biochemical recurrence occurred

in the UHF arm as defined by the Phoenix criterion. There was no significant difference in overall survival. Median follow-up for all groups combined was 55 months.

Conclusions: The UHF treatment scheme with HDR BB seems equivalent to standard treatment arms in terms of long-term toxicities and local control. The adoption of the UHF regimen would not only reduce the socioeconomic burden, it would offer the shortest treatment time for intermediate risk prostate cancer known to date, also of interest in the actual pandemic situation. Randomized control trials with larger cohorts are needed to further confirm our findings.

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OCULAR BRACHYTHERAPY FOR INTRA-OCULAR MELANOMA: FIVE-YEAR LARGE COHORT OUTCOMES IN THE PROVINCE OF ALBERTA

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Purpose: Intra-ocular melanoma was historically treated with surgical enucleation, which conferred morbidity from vision loss, pain, cosmetic and social stigma. The use of ocular brachytherapy to treat intraocular melanoma was established by the COMS (Collaborative Ocular Melanoma Study) in 2004. Here we present the Alberta experience in treating intraocular melanoma with brachytherapy in a large cohort with approximately five years follow up.

Materials and Methods: Patients with localized intra-ocular melanoma were treated between 2011 and 2020. All patients had at minimum ocular ultrasound and CT staging. Treatment was done with temporary ocular brachytherapy at the Cross Cancer Institute with I-125 high-dose seeds (4.14 mCi average activity) mounted in COMS or Eye Physics-brand (Los Alamitos, CA) plaque. The planned treatment volume (PTV) was covered by 100% of the prescription dose of 70Gy and included 2mm beyond tumour base and extended to apex. Dose to critical structures was minimized through conformal techniques including variable loading, variable seed strength, and asymmetric placement. Kaplan Meier estimate curves were generated, and cox proportional hazard regression was utilized to evaluate statistical significant differences. We stratified for survival by Gene Expression Profiling (GEP) class and the tumour's Largest Base Diameter (LBD), measured time from treatment to recurrence/metastasis/death.

Results: Three hundred eighty-nine patients (average age 67) were treated with a median follow-up of 4.5 years. Median dose achieved was 70.3Gy. The main organs-at-risk (OARs) were optic disc and fovea, which received a median dose of 37.6Gy and 47.9Gy, respectively. There were 62 cases of disease failure by way of distant metastases, with no local failures. Three patients underwent post-brachy enucleations due to treatment toxicity. At five years, overall progression-free survival (PFS) was 86.6%, and overall survival was 87.3% (most deaths were not melanoma-related). Five-year PFS for GEP Class 1A/1B was 93.1% versus 75.3% for Class 2 (HR 4.5, p<0.001). Five-year PFS for patients with LBD <12mm was 97.7% versus 83.4% for LBD ≥12mm (HR 7.7, p=0.001).

Conclusions: Ocular brachytherapy provides excellent local control and overall survival in treating intraocular uveal melanomas. Few serious post-RT toxicities were encountered requiring enucleation. LBD and GEP are strongly correlated with PFS. In Alberta, ocular brachytherapy continues to be a logistically feasible and effective organ-sparing approach to treating intra-ocular malignancy.